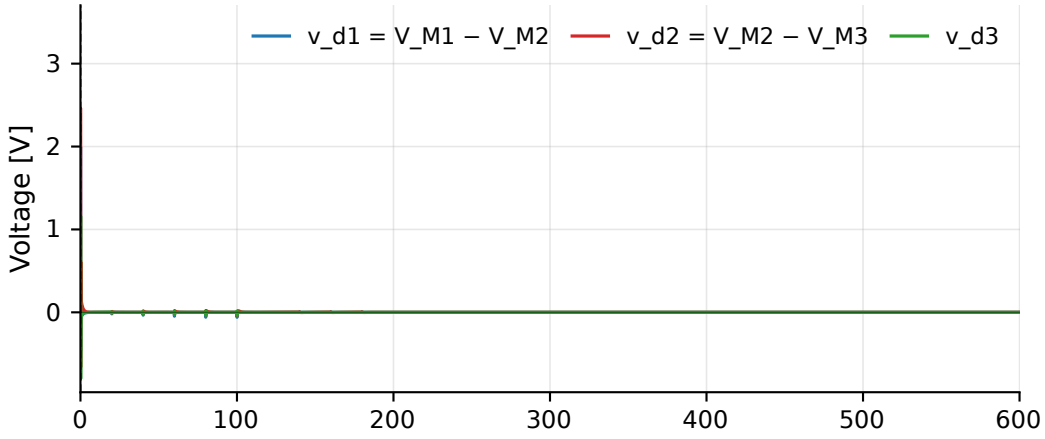
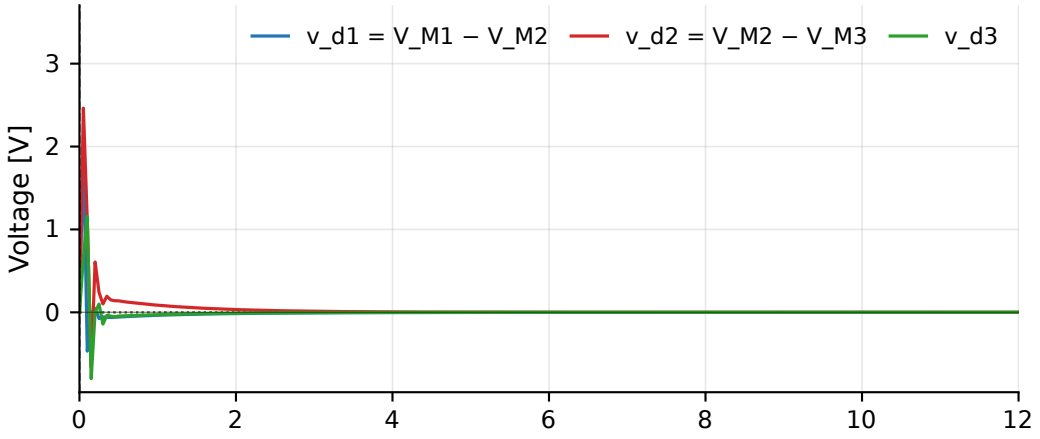
# MIL validate suite (4 modules): mppt-shade — last module 500 W/m², refs [0, 0, 0], mppt, PASS; cold start, kscale 1

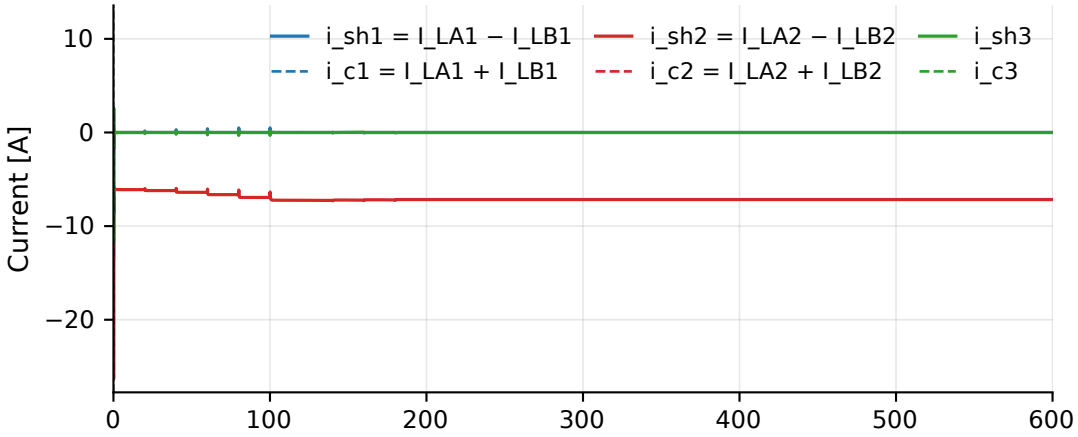
## Voltage differences v_d (full run); dotted = reference



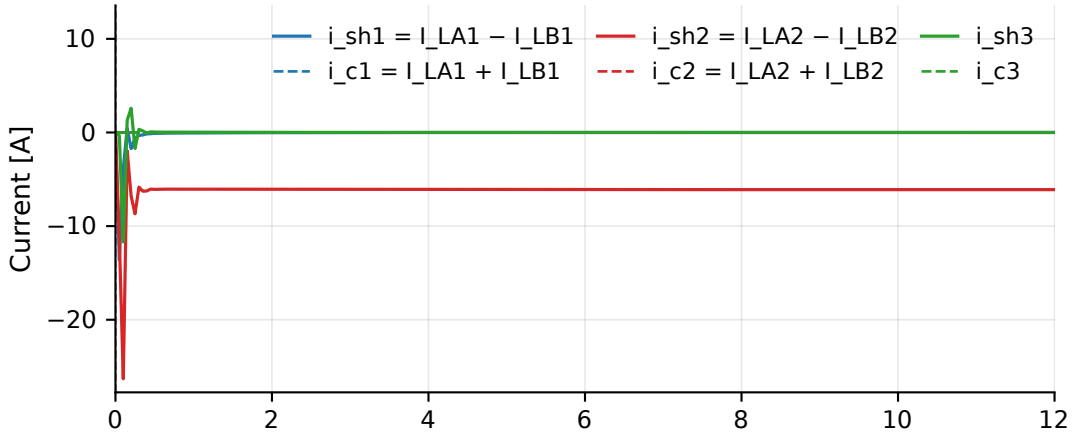

## Voltage differences v_d (zoom); dotted = reference



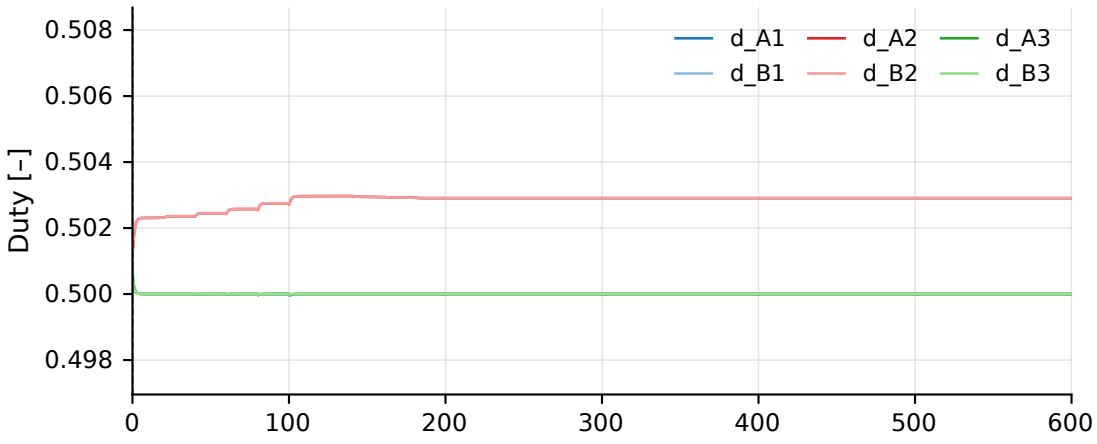

## Shuffled i_sh (solid) and circulating i_c (dashed) (full run)



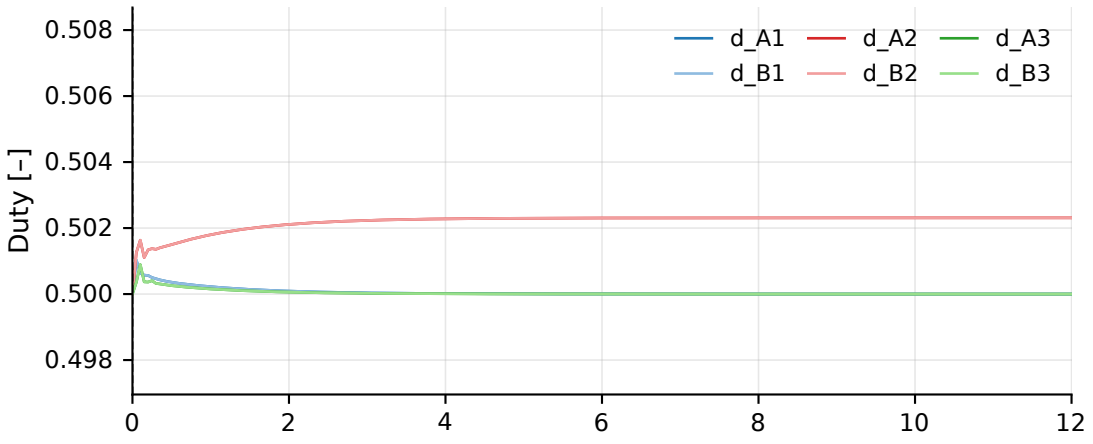

## Shuffled i_sh (solid) and circulating i_c (dashed) (zoom)



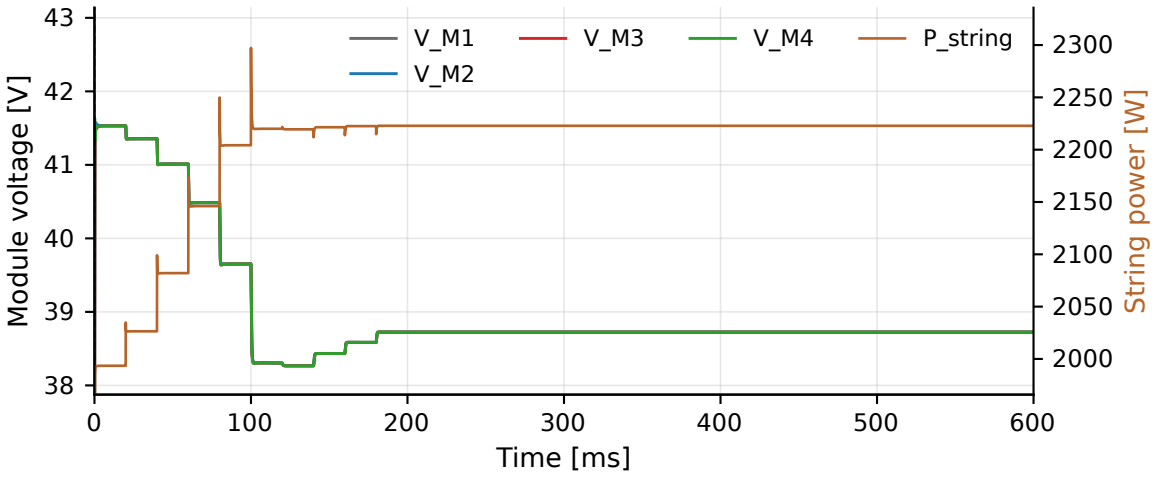

## Duty cycles (full run)

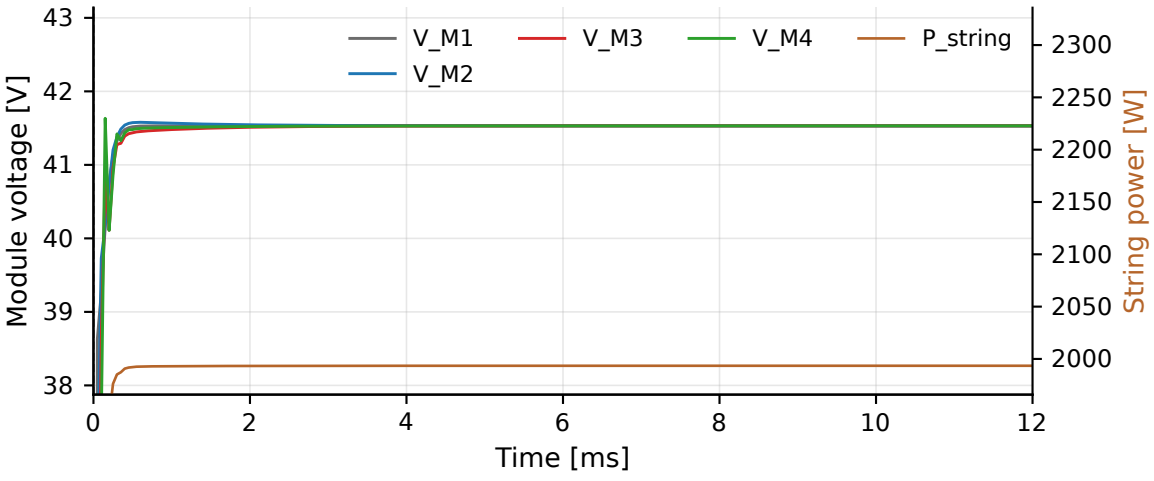



## Duty cycles (zoom)




## Module voltages and string power (full run)




## Module voltages and string power (zoom)